

CSCE 647 IMAGE SYNTHESIS · MODULE 13 · LECTURE NOTES

Spectral Rendering, Cameras, and Getting the Image Right

From radiance to pixels, and whether to believe them.

MODULE OUTCOMES

Explain when RGB rendering is wrong and what spectral rendering fixes.

Implement hero wavelength sampling.

Implement a thin-lens camera and explain the parameters photographically.

Explain tone mapping and colour management end to end.

Validate a renderer against something whose answer is known.

1 Spectral rendering

Light is a continuous function of wavelength. RGB rendering represents it with three numbers, and for most purposes that is adequate. These are the cases where it is not.

Case	Why RGB fails	Visibility
Dispersion	The refractive index varies with wavelength, so different wavelengths refract differently. With three channels you get three sharply separated bands.	Obvious and wrong-looking in prisms, diamonds, and thick glass edges.
Fluorescence	Energy absorbed at one wavelength is re-emitted at another. There is no RGB operation that moves energy between channels in this way.	Optical brighteners, safety clothing, some minerals.
Repeated multiplication	Multiplying RGB triples is not equivalent to multiplying spectra and converting afterwards — see below.	The case people do not expect. No glass required.
Thin films, structural colour	Interference depends on the wavelength directly.	Soap bubbles, oil films, beetle shells, CD surfaces.

WHY REPEATED MULTIPLICATION IS THE SURPRISING CASE

Converting a spectrum to RGB is a projection onto three basis functions. **Projection does not commute with multiplication.**

Concretely: light with spectrum S_1 reflecting off a surface with reflectance S_2 gives a result whose RGB is $\int S_1 S_2 x(\lambda) d\lambda$, which is *not* the product of the individual RGB projections. The approximation is exact only when one of the spectra is flat.

And the error compounds with every bounce. In a room with saturated coloured walls, the RGB answer diverges visibly from the spectral one after two or three interreflections — the colour bleeding is noticeably more saturated than it should be.

This is the case that surprises people, because it requires no glass, no exotic material, and no obviously spectral phenomenon. Just strongly coloured surfaces lit indirectly, which is an ordinary interior.

HERO WAVELENGTH SAMPLING

The naive approach assigns each path a single randomly chosen wavelength. This is correct, and it costs roughly four times as many paths to reach the same colour noise, because each path now carries information about only a slice of the spectrum.

Hero wavelength sampling (Wilkie et al., 2014) carries four wavelengths down the same path: one sampled 'hero' wavelength plus three more, rotated evenly through the visible range. The path geometry is shared; only the spectral quantities are evaluated four times.

The four are combined using multiple importance sampling — Module 08's machinery applied across wavelengths rather than across light-sampling strategies. Where the path was chosen based on the hero wavelength (at a dispersive interface, say), the hero gets the weight; elsewhere all four contribute equally.

The result nearly eliminates colour noise at almost no additional cost, because the expensive part of a sample is tracing the path, not evaluating four spectral values along it. It has been standard practice in spectral renderers since it was published.

2 Cameras

A pinhole camera is a useful default and is not what photographs come from. The **thin lens** model adds depth of field for the cost of two extra random numbers per ray: sample a point on the aperture disk, and aim the ray through the point where the pinhole ray would have crossed the focal plane.

$$\text{CoC} = A \cdot |s - f| / s \quad N = f / A$$

Parameter	Meaning	Why expose it this way
Focal length f	Field of view.	A photographer knows what 35 mm and 135 mm look like. 'Field of view in degrees' is less useful to the person operating it.
f-number N	Aperture size, as f/N .	$f/1.4$ is wide with shallow depth of field; $f/16$ is narrow and deep. Again, already understood.

Parameter	Meaning	Why expose it this way
Focus distance	Where the focal plane sits.	Usually better specified by picking a point in the scene than by typing a number.

- **Aperture shape.** Real apertures are polygons formed by the diaphragm blades, and out-of-focus highlights take that shape. Sampling a hexagon instead of a disk is a few lines and is most of what makes rendered bokeh look photographic.
- **Shutter interval.** Sample a time per ray within the exposure window; the scene is evaluated at that time. Motion blur then costs nothing beyond the transforms.
- **Rolling shutter.** Make the sampled time depend on the pixel's row, and you get the characteristic skew of a CMOS sensor on fast motion.
- **Reconstruction filter.** The pixel integral of Module 01 is weighted by a filter. **Use something other than a box** — Mitchell is the usual default, Gaussian is softer. A box filter visibly aliases, and switching costs nothing.

3 From radiance to pixels

Step	Operation	Symptom if wrong
Spectral → XYZ	Integrate the spectrum against the CIE standard observer functions.	Hues are systematically wrong, particularly for saturated colours.
XYZ → working RGB	A 3×3 matrix for the chosen primaries (sRGB, ACEScg, Rec.2020).	Saturated colours shift; the image does not match other tools.
Tone map	Compress unbounded HDR values into displayable range.	Highlights clip to white or wash out to grey.
Gamut map	Handle colours outside the display's gamut.	Bright saturated regions clip in one channel and shift hue.
Encode	Apply the display transfer function (sRGB, or a gamma curve).	The image looks washed out and too bright in the midtones.

THE TRANSFER FUNCTION AND TONE MAPPING ARE DIFFERENT THINGS

The transfer function — sRGB encoding, loosely called 'gamma' — exists because displays and human perception are not linear in intensity. It is **required**, it is **invertible**, and it discards no information. Not applying it is simply a bug.

Tone mapping compresses an unbounded dynamic range into what a display can show. It is **lossy**, it is a **creative choice**, and there is no uniquely correct operator.

Doing only the transfer function gives blown highlights: the rendered sun has a value of 10^5 and the display can show 1, so everything bright becomes a flat white region.

Doing only the tone map gives a washed-out, low-contrast image, because linear values were handed to a display expecting encoded ones. **This is the single most common error in a first renderer**, and it is worth recognising on sight: midtones too bright, the whole image looking slightly foggy.

Operator	Character	Use
Reinhard	Simple, never clips, desaturates highlights noticeably.	A reasonable default for debugging.
Filmic (Hable)	Film-like toe and shoulder; higher contrast.	Games, where a punchy look is wanted.
ACES	Industry standard; well-behaved highlight roll-off and a defined colour pipeline around it.	Film and most production work.
AgX	Recent; strong hue preservation as values approach white, so bright saturated lights do not shift.	Increasingly a default — Blender adopted it.

Pick one, use it consistently, and state which one when comparing images. A great deal of confusion in this course comes from comparing your render against a reference that used a different tone map and concluding the renderer is wrong. **Compare linear HDR values, not tone-mapped pixels**, when you are

measuring correctness — tone mapping is a display decision applied after the rendering is finished.

4 Validation, and what this course was about

Kind	Test	What it proves
Analytic	Hemisphere integrates to 2π . White furnace test. Constant environment on a diffuse sphere. A disk light over a diffuse plane.	The result is exactly right in cases where the right answer is known in closed form.
Statistical	χ^2 tests on every sampler. Convergence plots with slope -0.5 .	The samplers draw from the densities they claim, and the estimator is converging as theory says it must.
Comparative	Difference against PBRT on the same scene. The difference image must be structureless noise.	Agreement with an independent, heavily validated implementation.
Internal consistency	Different integrators must agree at convergence. Enabling MIS must not change the converged image , only the variance.	No single strategy is systematically wrong in a way the others share.
Physical	The Cornell box against its published measured radiometric data.	Agreement with reality — the original Cornell box was built and measured, not invented, which is precisely why it exists.

A RENDERER HAS A GROUND TRUTH, WHICH IS UNUSUAL

Most programs cannot be checked this thoroughly. There is no analytic answer for whether a compiler's optimisation was correct in general, or whether a user interface is right.

A path tracer converges to a defined quantity — the value of the rendering equation for the scene as described — and that value can be computed independently, bounded analytically in special cases, and compared against measurements of physical scenes.

This course has leaned on that throughout, and the habit is the most transferable thing in it: when a quantity has a ground truth, build the comparison before building the thing. Module 05 asked you to write the validation suite before the integrator, and everything since has used it.

The distinction that matters most is the one from Module 02: a *noise* problem, a *bias* problem, and a *model* problem look similar in an image and are fixed completely differently. Being able to tell them apart quickly is most of the practical skill this course teaches.

Where this leaves you: able to state what a pixel is an integral of and to estimate that integral correctly; able to diagnose a rendering failure by its appearance; and in possession of a renderer whose every image you can defend. **CSCE 649** supplies the motion worth rendering, **CSCE 748** takes the camera and image side far further, and **CSCE 650** adds the constraint of a frame budget with a person inside it.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
PBRT 4th ed. — Chapters 4.5 and 5, Colour and Cameras	Spectral representation, the CIE observer, colour spaces, the thin lens model, and the film and filter machinery.
Wilkie et al. — Hero Wavelength Spectral Sampling (2014, free)	The technique of §1, with the MIS derivation.
Cornell Box data and measurements	Geometry, measured spectral reflectances, and radiometric measurements. The physical validation of §4.
Troy Sobotka — AgX and colour management writing	The display transform of §3, with a clear account of why hue preservation in highlights matters.
Poynton — Frequently Asked Questions about Gamma (free)	The clearest explanation of the transfer function, and why calling it 'gamma' causes so much confusion.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement spectral rendering with a single wavelength per path. Render a prism and confirm you get a continuous spectrum rather than three bands.
- 2 Implement hero wavelength sampling with four wavelengths. Compare colour noise against single-wavelength sampling at equal time.
- 3 Construct the multiplication error of §1: a room with saturated coloured walls, rendered in RGB and spectrally. Report the difference after two, four, and eight bounces.
- 4 Implement a thin-lens camera. Render the same scene at $f/1.4$, $f/4$, and $f/16$ and confirm the depth of field behaves as a photographer would expect.
- 5 Implement a polygonal aperture and render out-of-focus point highlights. Confirm the bokeh takes the blade shape.
- 6 Implement box, tent, and Mitchell reconstruction filters. Compare on a high-contrast edge and report which aliases.
- 7 Implement the full colour pipeline of §3. Then remove the sRGB encode and document the result — recognising this on sight is worth the exercise.
- 8 Implement Reinhard, ACES, and AgX tone mapping. Render a scene with a very bright saturated light and compare how each handles the highlight.
- 9 Render the Cornell box and compare against the published measured radiometric data, not just against another renderer. Report the agreement.

- 10 Project 2 is now due.** Submit the renderer, the validation suite results, the convergence plots, and the portfolio of images with sample counts and render times stated.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name four cases where RGB rendering is wrong, and say which is the surprising one.
- 2 Why is multiplying RGB triples not the same as multiplying spectra?
- 3 What is hero wavelength sampling, and why is it nearly free?
- 4 How does a thin-lens camera work, and what three parameters should be exposed?
- 5 What determines bokeh shape, and why does it matter?
- 6 List the five steps from spectral radiance to a displayed pixel.
- 7 Distinguish the transfer function from tone mapping on three axes.
- 8 What does an image look like when the sRGB encode is omitted?
- 9 Why must correctness comparisons be made on linear values rather than tone-mapped ones?
- 10 Give the five kinds of validation and what each one proves.